Adrenal Gland Pathophysiology and Pharmacology

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Learning Objectives

1. Evaluate the epidemiology, risk factors, and signs and symptoms of hyperaldosteronism.
2. Describe the etiology, clinical features, pathogenesis, pathophysiology, prognosis and treatment options of hypercortisolism (Cushing syndrome).
3. Identify the causes and consequences of oversecretion of adrenal androgens; describe the basis for adrenal virilizing disorders.
4. Describe the etiology, pathogenesis and pathophysiology of adrenal insufficiency, including Addison disease and Waterhouse-Friderichsen syndrome.
5. Discuss the pharmacologic management of adrenal insufficiency.
6. Describe the mechanisms of action, therapeutic uses, and adverse side effects of glucocorticoids.
7. Discuss the therapeutic options for the management of Cushing syndrome with a focus on mechanism of action and side effects.

Case 1

A 52-year-old man with returns to clinic for follow-up of uncontrolled hypertension. His current blood pressure medications are hydrochlorothiazide, amlodipine, lisinopril, and hydralazine. He is adherent with his medications, a low-salt diet, and exercises regularly. He checks his blood pressure at home regularly, and it has been consistently elevated. In the clinic, his blood pressure is 175/110 mmHg. Review of the patient's chart shows intermittent hypokalemia for the past 4 years. His physical exam is normal.

Laboratory results

- Serum pH 7.55 (high)
- Serum potassium 3.2 mEq/L (low)
- Serum bicarbonate 34 mEq/L (high)
- Serum creatinine 0.9 mg/dL (normal)
- Plasma aldosterone concentration 65 ng/mL (high)
- Plasma renin activity 0.1 ng/mL (low)
- PAC/PRA ratio 650 (high)

Question

What is the most likely cause of the patient's hypertension?

A. Primary hypertension
B. Cushing syndrome
C. Pheochromocytoma
D. Primary hyperaldosteronism
E. Renal artery stenosis

Question

Which of the following will be most appropriate to treat hypertension in this patient?

A. Spironolactone

B. Atorvastatin

C. Nitroglycerin

D. Furosemide

E. Metoprolol

Case 2

A 42-year-old woman with hypertension and elevated BMI presents with difficulty losing weight. Despite calorie restriction and increased activity, she has had a 10-pound weight gain over the last 6 months. She also reports polyuria, polydipsia, several vaginal yeast infections, easy bruising, and increased facial hair and acne. On exam, her blood pressure is 160/100 mmHg and she has central obesity with thin extremities and a rounded plethoric face. Posterior cervical and supraclavicular fat pads are enlarged, and wide violaceous striae are visible along the abdomen and axilla. Proximal muscle strength is decreased.

Laboratory Results and Question

Laboratory results

- 24–hr urine free cortisol: high
- ACTH level: high
- Low-dose dexamethasone suppression test: ACTH not suppressed
- High-dose dexamethasone suppression test: ACTH suppressed

What is the most likely cause of the patient's clinical presentation?

A. **ACTH-producing pituitary tumor**
B. ACTH-producing small cell lung carcinoma
C. Cortisol-producing adrenal cortical adenoma
D. Exogenous steroids

Question

What is the most appropriate pharmacotherapy for this patient?

A. Inhibitor of cortisol biosynthesis

B. Antagonist of mineralocorticoid receptors

C. Agonist of glucocorticoid receptors

D. Antagonist of adrenergic receptors

E. Inhibitor of catecholamine reuptake

Cushing Syndrome Etiologies

Exogenous corticosteroids

- Most common cause of Cushing syndrome
- Exclude before obtaining lab tests

ACTH-producing pituitary tumor

- Microadenoma most common endogenous cause (65-70% of cases)

Adrenocortical tumor

- Adenomas and carcinomas (15-20%)

Ectopic ACTH-production (non-pituitary tumor)

- Most common is small cell lung carcinoma (10-15% of cases)

Cushing syndrome, laboratory tests

Diagnosis	Serum or saliva cortisol, midnight	24-hour urine free cortisol	Low dose dexamethasone suppression	High dose dexamethasone suppression	ACTH
Normal	N	N	S	S	N
ACTH-producing pituitary tumor					
Ectopic ACTH					
Adrenal tumor					

ACTH = adrenocorticotropic hormone; N = normal; NS = not suppressed; S = suppressed; N/A = not applicable

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What would the tests for an ACTH-producing pituitary tumor show?

Diagnosis	Serum or saliva cortisol, midnight	24-hour urine free cortisol	Low dose dexamethasone suppression	High dose dexamethasone suppression	ACTH
Normal	N	N	S	S	N
ACTH-producing pituitary tumor					
Ectopic ACTH					
Adrenal tumor					

ACTH = adrenocorticotropic hormone; N = normal; NS = not suppressed; S = suppressed; N/A = not applicable

What would the tests for an ACTH-producing pituitary tumor show?

Diagnosis	Serum or saliva cortisol, midnight	24-hour urine free cortisol	Low dose dexamethasone suppression	High dose dexamethasone suppression	ACTH
Normal	N	N	S	S	N
ACTH-producing pituitary tumor	High	High	NS	S	High
Ectopic ACTH					
Adrenal tumor					

ACTH = adrenocorticotropic hormone; N = normal; NS = not suppressed; S = suppressed; N/A = not applicable

What would the tests for ectopic ACTH show?

Diagnosis	Serum or saliva cortisol, midnight	24-hour urine free cortisol	Low dose dexamethasone suppression	High dose dexamethasone suppression	ACTH
Normal	N	N	S	S	N
ACTH-producing pituitary tumor	High	High	NS	S	High
Ectopic ACTH					
Adrenal tumor					

ACTH = adrenocorticotropic hormone; N = normal; NS = not suppressed; S = suppressed; N/A = not applicable

What would the tests for ectopic ACTH show?

Diagnosis	Serum or saliva cortisol, midnight	24-hour urine free cortisol	Low dose dexamethasone suppression	High dose dexamethasone suppression	ACTH
Normal	N	N	S	S	N
ACTH-producing pituitary tumor	High	High	NS	S	High
Ectopic ACTH	High	High	NS	NS	High
Adrenal tumor					

ACTH = adrenocorticotropic hormone; N = normal; NS = not suppressed; S = suppressed; N/A = not applicable

What would the tests for an adrenal tumor show?

Diagnosis	Serum or saliva cortisol, midnight	24-hour urine free cortisol	Low dose dexamethasone suppression	High dose dexamethasone suppression	ACTH
Normal	N	N	S	S	N
ACTH-producing pituitary tumor	High	High	NS	S	High
Ectopic ACTH	High	High	NS	NS	High
Adrenal tumor					

ACTH = adrenocorticotropic hormone; N = normal; NS = not suppressed; S = suppressed; N/A = not applicable

What would the tests for an adrenal tumor show?

Diagnosis	Serum or saliva cortisol, midnight	24-hour urine free cortisol	Low dose dexamethasone suppression	High dose dexamethasone suppression	ACTH
Normal	N	N	S	S	N
ACTH-producing pituitary tumor	High	High	NS	S	High
Ectopic ACTH	High	High	NS	NS	High
Adrenal tumor	High	High	N/A	N/A	Low

ACTH = adrenocorticotropic hormone; N = normal; NS = not suppressed; S = suppressed; N/A = not applicable

Case 3

A 45-year-old woman presents with fatigue, lightheadedness, and anorexia with occasional nausea and vomiting for several months. She has unintentional weight loss and cravings for salty/sweet foods. On exam, blood pressure is 112/78 mmHg supine and 96/58 mmHg standing. She has hyperpigmentation on her face, in her palmar creases, and in the pattern of the bra straps.

Question

Laboratory results

- 24–hr urine free cortisol: low
- ACTH level: high
- Aldosterone level: low
- Renin level: high

What is the most likely cause of the patient's clinical presentation?

A. Primary adrenal insufficiency
B. Secondary adrenal insufficiency
C. Tertiary adrenal insufficiency

Question

What is the most appropriate treatment for this patient?

A. Glucocorticoid receptor agonist
B. Mineralocorticoid receptor agonist
C. Glucocorticoid and mineralocorticoid receptor agonists
D. Glucocorticoid receptor antagonist
E. Mineralocorticoid receptor antagonist
F. Both glucocorticoid and mineralocorticoid antagonists

Adrenal Insufficiency

Primary: Adrenal disease

- **Destruction of the adrenal glands**
  - Autoimmunity (Addison disease)
  - Metastatic tumor
  - Infection (TB)
  - Hemorrhage
  - Infiltrative disease (hemochromatosis, sarcoidosis)

Secondary: Pituitary disease (decreased ACTH)

- **Destruction of the pituitary corticotroph cells**
  - Tumor
  - Trauma
  - Hemorrhage
  - Infiltrative disease (hemochromatosis, sarcoidosis)

Tertiary: Hypothalamic disease

- **Destruction of the paraventricular nucleus of the hypothalamus**
  - Tumor
  - Trauma
  - Infection

Adrenal insufficiency, laboratory tests

Diagnosis	Site of insufficiency	24-hr urine cortisol	ACTH	CRH	Aldosterone	Renin
Normal	N/A	Normal	Normal	Normal	Normal	Normal
Primary	Adrenal					
Secondary	Pituitary					
Tertiary	Hypothalamus					

ACTH = adrenocorticotropic hormone; CRH = corticotropin-releasing hormone

What will the tests for PRIMARY adrenal insufficiency show?

Diagnosis	Site of insufficiency	24-hr urine cortisol	ACTH	CRH	Aldosterone	Renin
Normal	N/A	Normal	Normal	Normal	Normal	Normal
Primary	Adrenal					
Secondary	Pituitary					
Tertiary	Hypothalamus					

ACTH = adrenocorticotropic hormone; CRH = corticotropin-releasing hormone

What will the tests for PRIMARY adrenal insufficiency show?

Diagnosis	Site of insufficiency	24-hr urine cortisol	ACTH	CRH	Aldosterone	Renin
Normal	N/A	Normal	Normal	Normal	Normal	Normal
Primary	Adrenal	Low	High	High	Low	High
Secondary	Pituitary					
Tertiary	Hypothalamus					

ACTH = adrenocorticotropic hormone; CRH = corticotropin-releasing hormone

What will the tests for SECONDARY adrenal insufficiency show?

Diagnosis	Site of insufficiency	24-hr urine cortisol	ACTH	CRH	Aldosterone	Renin
Normal	N/A	Normal	Normal	Normal	Normal	Normal
Primary	Adrenal	Low	High	High	Low	High
Secondary	Pituitary					
Tertiary	Hypothalamus					

ACTH = adrenocorticotropic hormone; CRH = corticotropin-releasing hormone

What will the tests for SECONDARY adrenal insufficiency show?

Diagnosis	Site of insufficiency	24-hr urine cortisol	ACTH	CRH	Aldosterone	Renin
Normal	N/A	Normal	Normal	Normal	Normal	Normal
Primary	Adrenal	Low	High	High	Low	High
Secondary	Pituitary	Low	Low	High	Normal or low	High
Tertiary	Hypothalamus					

ACTH = adrenocorticotropic hormone; CRH = corticotropin-releasing hormone

What will the tests for TERTIARY adrenal insufficiency show?

Diagnosis	Site of insufficiency	24-hr urine cortisol	ACTH	CRH	Aldosterone	Renin
Normal	N/A	Normal	Normal	Normal	Normal	Normal
Primary	Adrenal	Low	High	High	Low	High
Secondary	Pituitary	Low	Low	High	Normal or low	High
Tertiary	Hypothalamus					

ACTH = adrenocorticotropic hormone; CRH = corticotropin-releasing hormone

What will the tests for TERTIARY adrenal insufficiency show?

Diagnosis	**Site of insufficiency**	**24-hr urine cortisol**	**ACTH**	**CRH**	**Aldosterone**	**Renin**
Normal	N/A	Normal	Normal	Normal	Normal	Normal
Primary	Adrenal	Low	High	High	Low	High
Secondary	Pituitary	Low	Low	High	Normal or low	High
Tertiary	Hypothalamus	Low	Low	Low	Normal or low	High

ACTH = adrenocorticotropic hormone; CRH = corticotropin-releasing hormone

Case 4

An 8-day-old girl is brought for evaluation of recurrent vomiting, poor feeding, and irritability. Her parents are concerned that she has been sleeping too much and that she does not wake up to feed on her own. Heart rate is elevated and blood pressure is low. On physical examination, she is lethargic with dry mucous membranes and a sunken anterior fontanelle. There is a markedly enlarged clitoris

Question

Laboratory results

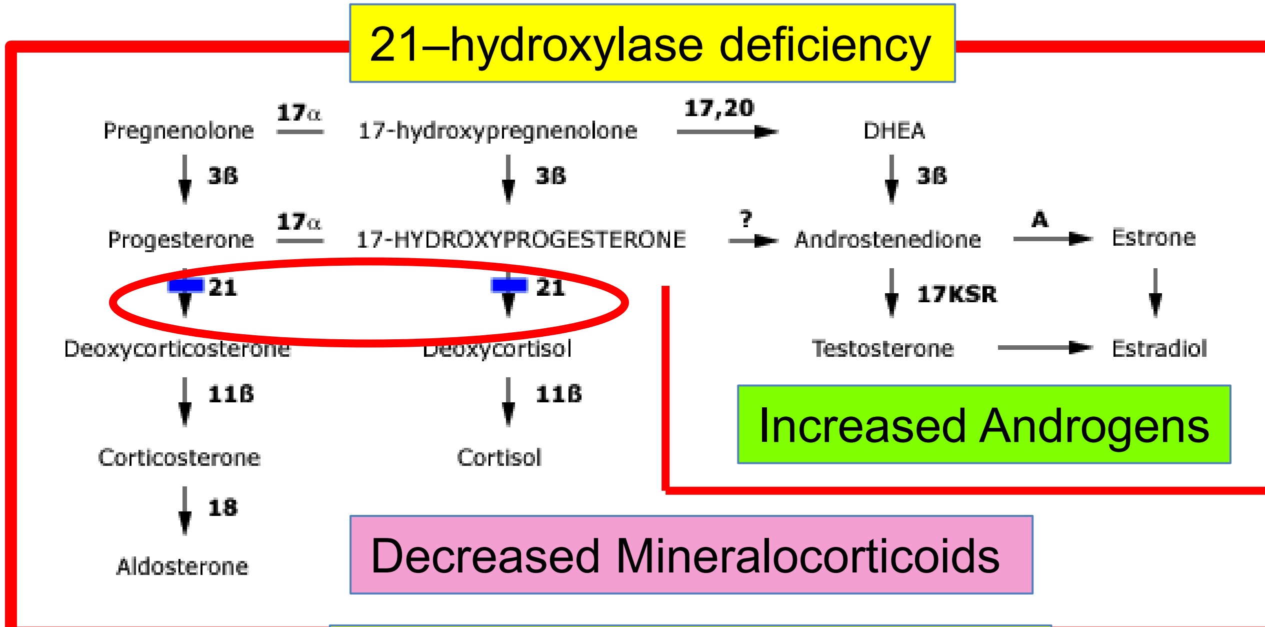
- Serum sodium: low
- Serum potassium: high
- Serum aldosterone: low
- Serum testosterone: high

What is the most likely cause of the patient's clinical presentation?

A. 21-hydroxylase deficiency
B. 11-hydroxylase deficiency
C. 17-hydroxylase deficiency

Adrenogenital Syndromes

- Pathophysiology
  - Defects in glucocorticoid synthesis resulting in increased androgens and increased/decreased mineralocorticoids
  - Pituitary ACTH increases–>adrenal hyperplasia
  - Due to inborn errors of metabolism
    - All have **DECREASED** cortisol
    - 21–hydroxylase deficiency
      - **DECREASED** mineralocorticoids, **INCREASED** androgens
    - 11–beta–hydroxylase deficiency
      - **INCREASED** mineralocorticoids, **INCREASED** androgens
    - 17–hydroxylase deficiency
      - **INCREASED** mineralocorticoids, **DECREASED** androgens
- Clinical Presentation
  - Genitalia: Clitoromegaly (21-OH, 11-OH), micropenis (17-OH)
  - BO: Hypotension (21-OH) or hypertension (11-OH, 17-OH)
  - Diagnose by testing for specific steroid excess
    - 21–OH: 17–hydroxyprogesterone
    - 11–OH: deoxycorticosterone, deoxycortisol
    - 17–OH: deoxycorticosterone, corticosterone
- Treatment
  - Corticosteroids
  - ± mineralocorticoids for 21–hydroxylase deficiency


21–hydroxylase deficiency
Pregnenolone
17α
17-hydroxypregnenolone
17,20
DHEA
3β
3β
3β
Progesterone
17α
17-HYDROXYPROGESTERONE
?
Androstenedione
A
Estrone
21
21
17KSR
Deoxycorticosterone
Deoxycortisol
Testosterone
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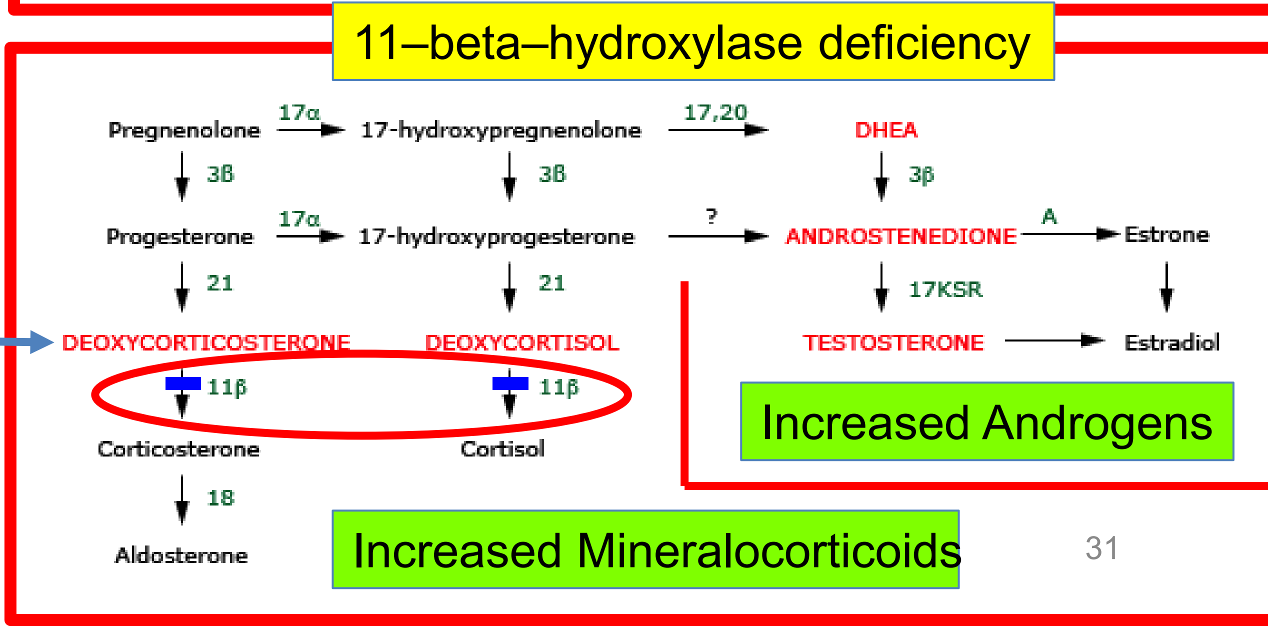
11β
Increased Androgens
Corticosterone
Cortisol
18
Aldosterone
Decreased Mineralocorticoids


11–beta–hydroxylase deficiency
Pregnenolone
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3β
Progesterone
17α
17-hydroxyprogesterone
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ANDROSTENEDIONE
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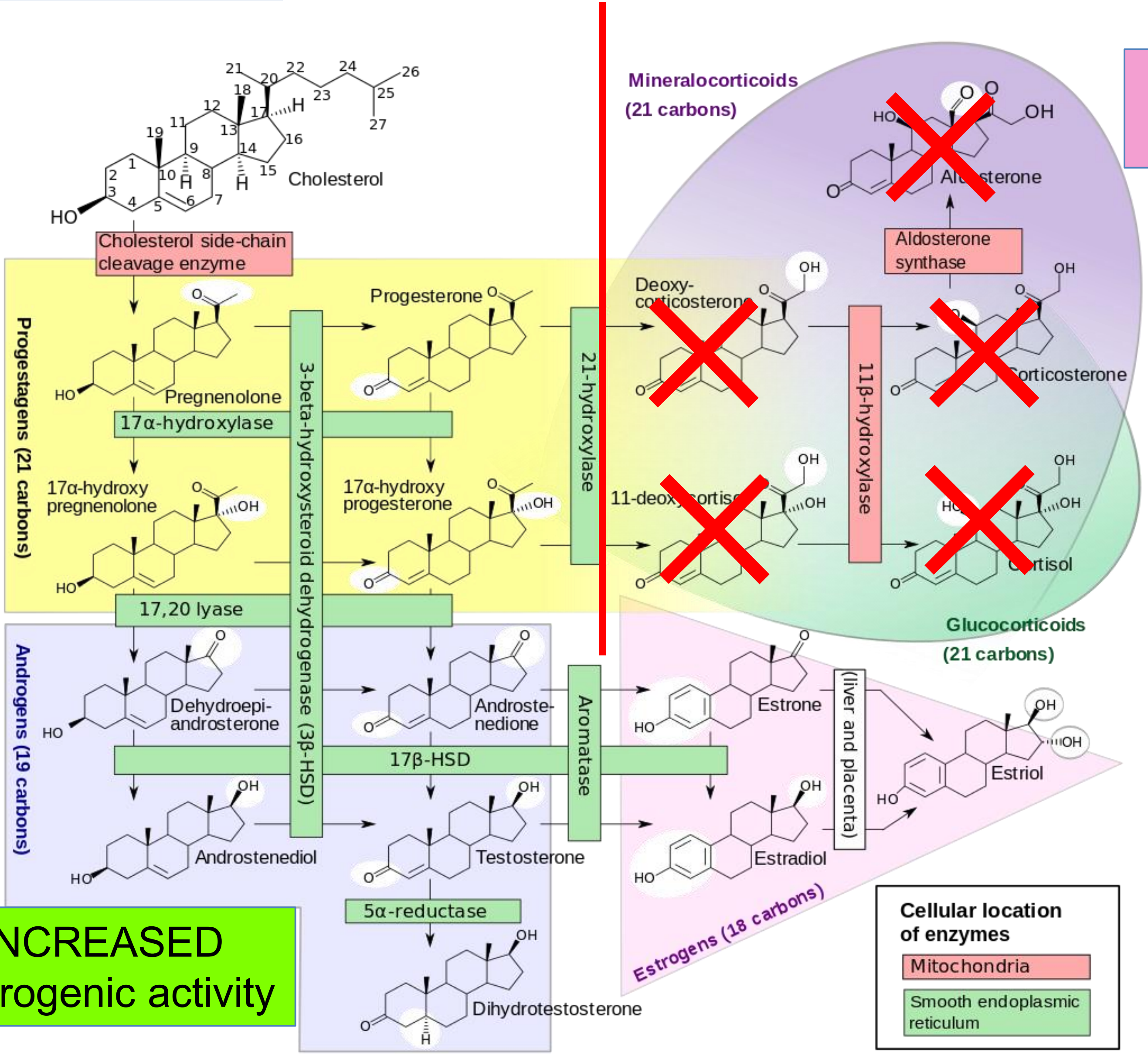
Mineralocorticoid activity
DEOXYCORTICOSTERONE
DEOXYCORTISOL
TESTOSTERONE
Estradiol
11β
11β
Increased Androgens
Corticosterone
Cortisol
18
Aldosterone
Increased Mineralocorticoids

21-hydroxylase deficiency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Cholesterol side-chain cleavage enzyme
Progestagens (21 carbons)
Pregnenolone
HO
O
17α-hydroxylase
17α-hydroxy pregnenolone
OH
3-beta-hydroxysteroid dehydrogenase (3β-HSD)
Progesterone
17α-hydroxy progesterone
21-hydroxylase
17,20 lyase
Androgens (19 carbons)
Dehydroepi-androsterone
Androste-nedione
17β-HSD
Androstenediol
Testosterone
5α-reductase
Dihydrotestosterone
Aromatase
Mineralocorticoids (21 carbons)
Deoxy-corticosterone
11-deoxycortisol
11β-hydroxylase
Corticosterone
Cortisol
Aldosterone synthase
Aldosterone
Glucocorticoids (21 carbons)
Estrone
Estradiol
Estriol
(liver and placenta)
Estrogens (18 carbons)
Cellular location of enzymes
Mitochondria
Smooth endoplasmic reticulum

DECREASED mineralocorticoid activity

DECREASED glucocorticoid activity

INCREASED androgenic activity

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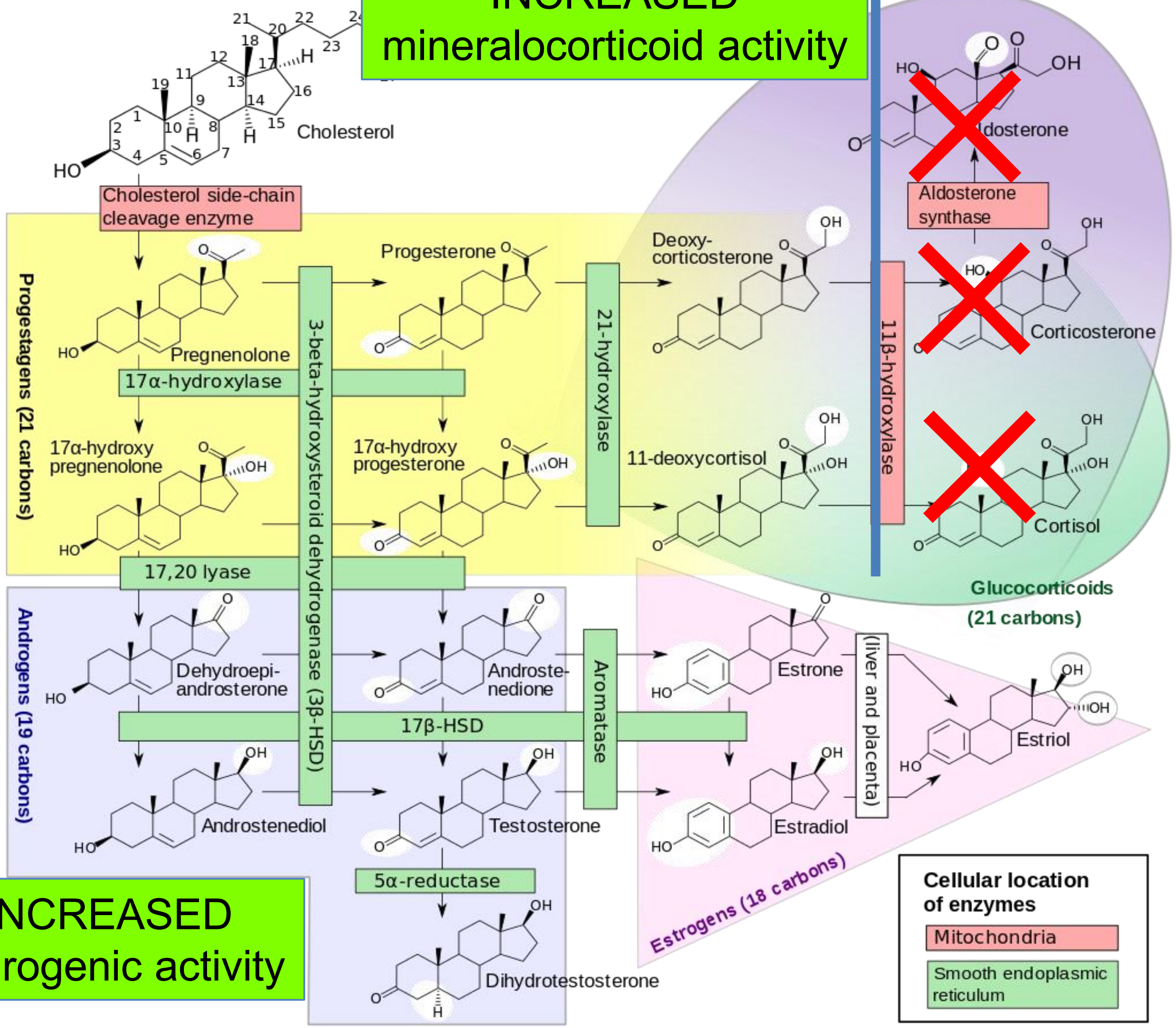
11-hydroxylase deficiency deficiency


Cholesterol
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Cholesterol side-chain cleavage enzyme
Progestagens (21 carbons)
Pregnenolone
17α-hydroxylase
17α-hydroxy pregnenolone
17,20 lyase
3-beta-hydroxysteroid dehydrogenase (3β-HSD)
Progesterone
17α-hydroxy progesterone
21-hydroxylase
Deoxy-corticosterone
11-deoxycortisol
11β-hydroxylase
Aldosterone
Aldosterone synthase
Corticosterone
Cortisol
Glucocorticoids (21 carbons)
Androgens (19 carbons)
Dehydroepi-androsterone
Androstenediol
Androste-nedione
Testosterone
17β-HSD
5α-reductase
Dihydrotestosterone
Aromatase
Estrone
Estradiol
(liver and placenta)
Estriol
Estrogens (18 carbons)
Cellular location of enzymes
Mitochondria
Smooth endoplasmic reticulum
OH
O

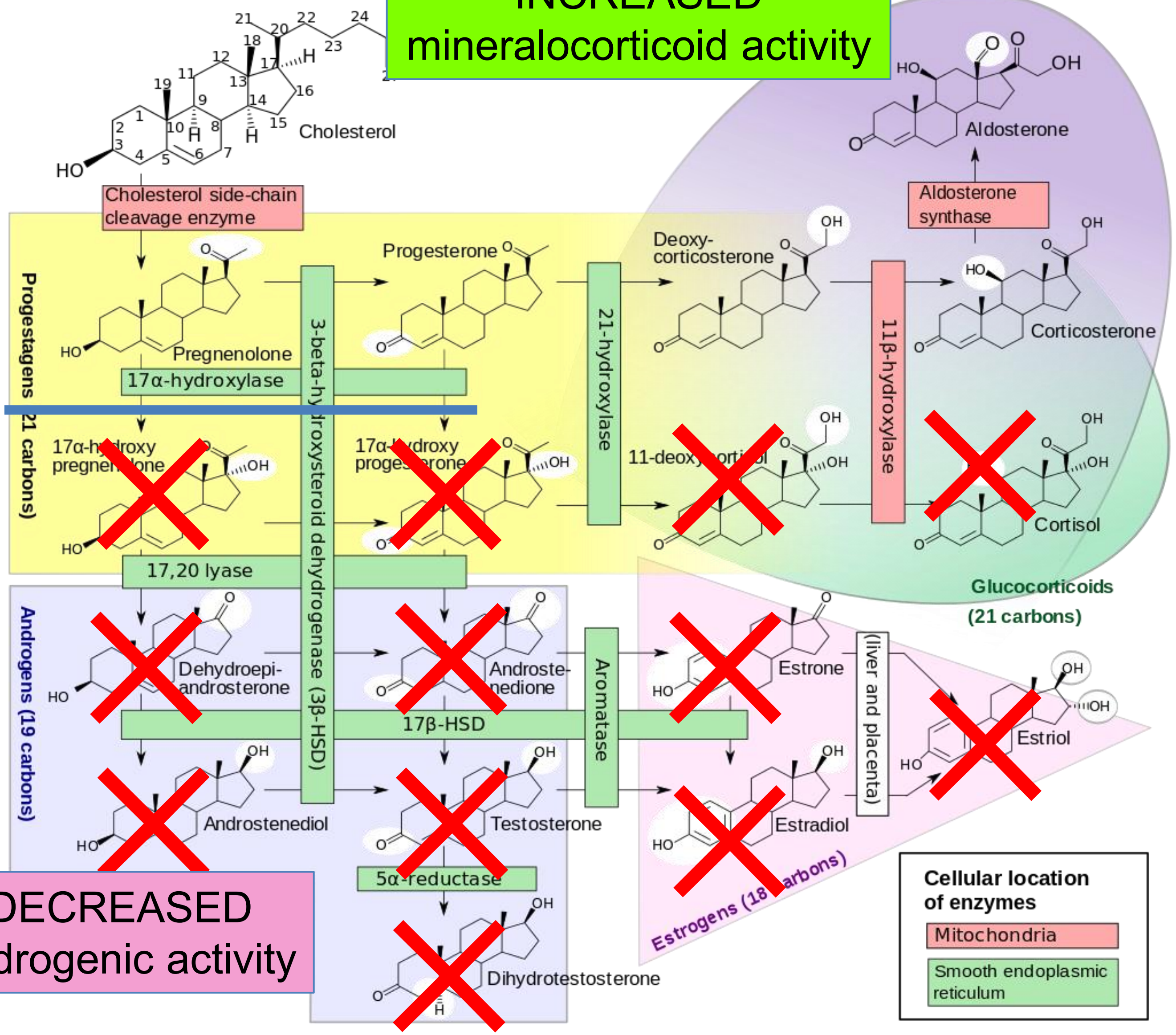
INCREASED mineralocorticoid activity

DECREASED glucocorticoid activity

INCREASED androgenic activity

17-hydroxylase deficiency deficiency

INCREASED mineralocorticoid activity


Cholesterol
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HO
Cholesterol side-chain cleavage enzyme
Progestagens (21 carbons)
Pregnenolone
17α-hydroxylase
3-beta-hydroxysteroid dehydrogenase (3β-HSD)
Progesterone
21-hydroxylase
Deoxy-corticosterone
11β-hydroxylase
Corticosterone
Aldosterone synthase
Aldosterone
17α-hydroxy pregnenolone
17α-hydroxy progesterone
11-deoxycortisol
Cortisol
Glucocorticoids (21 carbons)
17,20 lyase
Androgens (19 carbons)
Dehydroepi-androsterone
Androste-nedione
Aromatase
Estrone
(liver and placenta)
Estriol
17β-HSD
Androstenediol
Testosterone
Estradiol
5α-reductase
Dihydrotestosterone
Estrogens (18 carbons)
OH
O
Cellular location of enzymes
Mitochondria
Smooth endoplasmic reticulum

DECREASED glucocorticoid activity

DECREASED androgenic activity

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	21–OH	11–OH	17–OH
Incidence	1:20,000	1:200,000	1:1,000,000
Mineralocorticoids[1]	Low	High	High
Serum Na	Low	High	High
Serum K	High	Low	Low
Androgens[2]	High	High	Low
Genitalia changes (M/F)	None/ clitoromegaly	None/ clitoromegaly	Micropenis/ none
Diagnostic test	17-OH P	DOC, deoxycortisol	DOC, corticosterone
Treatment	GC ± MC	GC	GC

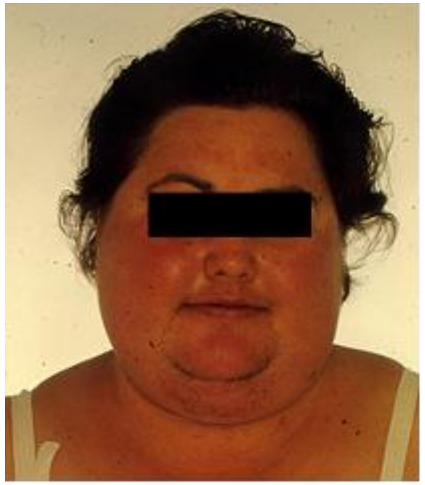
17-OH P = 17-hydroxyprogesterone, DOC = deoxycorticosterone, GC = glucocorticoids, MC = mineralocorticoids, M = male, F = female

[1] Determines changes in blood pressure, Na, and K. [2] Determines changes in genitalia.

Case 5

A 41-year-old woman with rheumatoid arthritis (RA) presents for evaluation of weight gain. She was diagnosed 6 months ago and is on 20 mg of prednisone per day to control her symptoms. She has a 30-pound weight gain, increased facial hair, acne, frequent urination, and thirst. Her blood pressure is 150/92 mmHg. Exam reveals central obesity, cervical fat pad enlargement, moon facies, dark purple abdominal striae, and decreased proximal muscle strength.

Representative Physical Findings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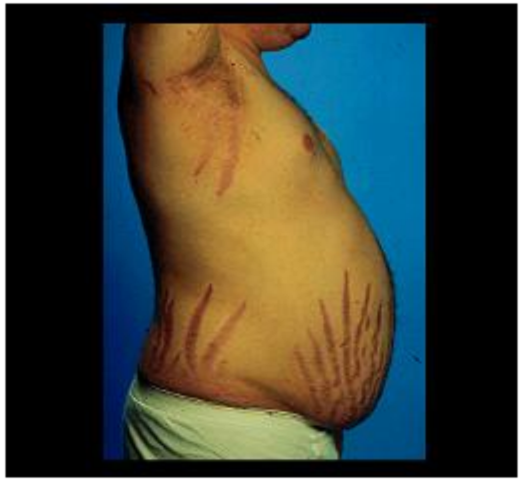
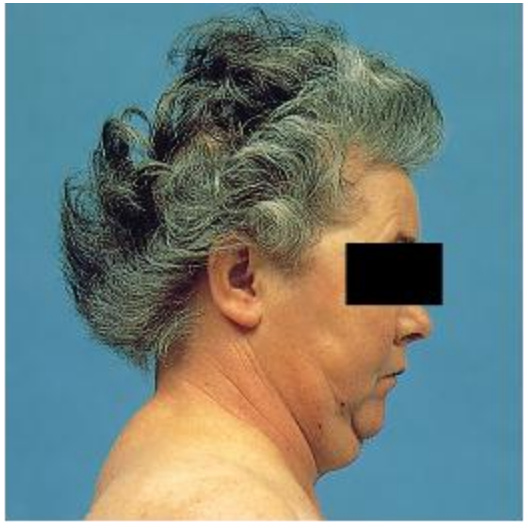
Moon facies



Purple striae and
central obesity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Cervical fat pad
enlargement

Question

Which of the following laboratory abnormality is most likely from her current medication?

A. Hypochloremia
B. Hypokalemia
C. Hypoglycemia
D. Hyponatremia

What would the tests for exogenous glucocorticoids show?

Diagnosis	Serum or saliva cortisol, midnight	24-hour urine free cortisol	Low dose dexamethasone suppression	High dose dexamethasone suppression	ACTH
Normal	N	N	S	S	N
ACTH-producing pituitary tumor	High	High	NS	S	High
Ectopic ACTH	High	High	NS	NS	High
Adrenal tumor	High	High	N/A	N/A	Low
Exogenous GC					

ACTH = adrenocorticotropic hormone; N = normal; GC = glucocorticoids; NS = not suppressed;
S = suppressed; N/A = not applicable

What would the tests for exogenous glucocorticoids show?

Diagnosis	Serum or saliva cortisol, midnight	24-hour urine free cortisol	Low dose dexamethasone suppression	High dose dexamethasone suppression	ACTH
Normal	N	N	S	S	N
ACTH-producing pituitary tumor	High	High	NS	S	High
Ectopic ACTH	High	High	NS	NS	High
Adrenal tumor	High	High	N/A	N/A	Low
Exogenous GC	High	High	N/A	N/A	Low

ACTH = adrenocorticotropic hormone; N = normal; GC = glucocorticoids; NS = not suppressed; S = suppressed; N/A = not applicable

Answer slides

Answer: D

What is the most likely cause of the patient's hypertension?

A. Cushing syndrome

B. Primary hypertension

C. Pheochromocytoma

D. Primary hyperaldosteronism

E. Renal artery stenosis

Answer A

Which of the following will be most appropriate to treat hypertension in this patient?

A. Spironolactone

B. Atorvastatin

C. Nitroglycerin

D. Furosemide

E. Metoprolol

Answer: A

Laboratory results

- 24–hr urine free cortisol: high
- ACTH level: high
- Low-dose dexamethasone suppression test: ACTH not suppressed
- High-dose dexamethasone suppression test: ACTH suppressed

What is the most likely cause of the patient's clinical presentation?

A. ACTH-producing pituitary tumor

B. ACTH-producing small cell lung carcinoma

C. Cortisol-producing adrenal cortical adenoma

D. Exogenous steroids

Answer: A

What is the most appropriate pharmacotherapy for this patient?

A. Inhibitor of cortisol biosynthesis

B. Antagonist of mineralocorticoid receptors

C. Agonist of glucocorticoid receptors

D. Antagonist of adrenergic receptors

E. Inhibitor of catecholamine reuptake

Answer: A

Laboratory results

- 24–hr urine free cortisol: low
- ACTH level: high
- CRH level: high
- Aldosterone level: low
- Renin level: high

What is the most likely cause of the patient's clinical presentation?

A. Primary adrenal insufficiency

B. Secondary adrenal insufficiency

C. Tertiary adrenal insufficiency

Answer: B

What is the most appropriate treatment for this patient?

A. Glucocorticoid receptor agonist
B. Mineralocorticoid receptor agonist
C. Both glucocorticoid and mineralocorticoid recpeptor agonists
D. Glucocorticoid receptor antagonist
E. Mineralocorticoid receptor antagonist
F. Both glucocorticoid and mineralocorticoid antagonists

Answer : B

Laboratory results

- Sodium: low
- Potassium: high
- Aldosterone: low
- Testosterone: high

What is the most likely cause of the patient's clinical presentation?

A. 21-hydroxylase deficiency

B. 11-hydroxylase deficiency

C. 17-hydroxylase deficiency

Answer: B

Which of the following side effect is most likely in this patient from her current medication?

A. Hypochloremia

B. Hypokalemia

C. Hypoglycemia

D. Hyponatremia